

- Marginals

$$P_X(x) = \sum_y P_{XY}(x, y) \quad , \quad P_Y(y) = \sum_x P_{XY}(x, y)$$

- Independence

$$P_{XY}(x, y) = P_X(x) \cdot P_Y(y)$$

$$F_{XY}(x, y) = F_X(x) \cdot F_Y(y)$$

both must be true

- $E[XY]$

expected value of the product

$$\begin{aligned} E[X] &= \sum_x x \cdot P_X(x) \\ &= \sum_x \sum_y x \cdot P_{XY}(x, y) \end{aligned} \quad \left| \quad \begin{aligned} E[Y] &= \sum_y y \cdot P_Y(y) \\ &= \sum_y \sum_x y \cdot P_{XY}(x, y) \end{aligned}$$

$$E[XY] = \sum_x \sum_y xy \cdot P_{XY}(x, y)$$

If X and Y are independent,	$P_{XY}(x, y) = P_X(x) \cdot P_Y(y)$ $E[XY] = E[X] \cdot E[Y]$ $F_{XY}(x, y) = F_X(x) \cdot F_Y(y)$
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- Consistency conditions

$$\sum_{x, y} P_{XY}(x, y) = 1$$

$$F_{XY}(\infty, \infty) = 1$$

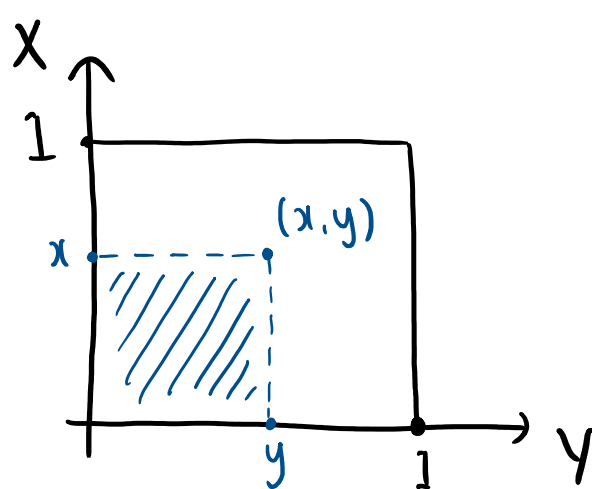
$$F_{XY}(-\infty, -\infty) = 0$$

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$$\left. \begin{aligned} F_{XY}(x, \infty) &= F_X(x) \\ F_{XY}(\infty, y) &= F_Y(y) \end{aligned} \right\} \text{ marginal CDF}$$

- Multiple continuous random variables

e.g. picking a number uniformly at random from a unit square centered at $(0.5, 0.5)$



$$F_{XY}(x, y) = xy \quad (\text{area})$$

$$\frac{\partial F_{XY}(x, y)}{\partial x} = \lim_{h \rightarrow 0} \frac{F_{XY}(x+h, y) - F_{XY}(x, y)}{h}$$

rate of change of F_{XY} in x -direction.

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y} \Rightarrow \text{JOINT PDF}$$

$\int_{xy} f(x, y) \cdot dx \cdot dy \rightarrow$ probability that (X, Y) are in a rectangle of area of $dx dy$ around (x, y) .

$$F_{XY}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{XY}(s, t) \cdot ds dt$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$